# SSLANet: A Frequency–Spatial Visual System for Efficient Hyperspectral Image Classification

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***Abstract*—Hyperspectral image classification requires a compact representation that can jointly capture discriminative spectral cues, local spatial structures, and efficient inference behavior under limited annotations. Existing CNN-based methods provide effective local pattern extraction but are less capable of modeling long-range spectral dependencies. Transformer-based models improve global interaction at the cost of considerable computational overhead, whereas recent state-space models usually formulate hyperspectral data as spatial–spectral sequences and still leave transform-domain spectral variations insufficiently explored. To address these limitations, we propose a spectral–spatial learning adaptive network (SSLANet), an efficient frequency–spatial learning network for hyperspectral image classification. The core component is a residual frequency–spatial layer that integrates an adaptive spectral block (ASB) and an interactive convolution block (ICB). ASB performs learnable complex-valued spectral modulation together with input-dependent frequency-energy selection to enhance material-sensitive spectral responses, while ICB propagates the filtered features into local neighborhoods through multiplicative spatial–channel interaction. This design differs from fixed transform-domain preprocessing in that the frequency response is conditioned on the current input feature distribution through a soft mask and an effective transfer function in the latent frequency domain. Extensive experiments on three public hyperspectral benchmarks demonstrate that SSLANet achieves competitive classification accuracy with favorable computational efficiency. The source code is available at https://github.com/NELUJA86/SSLANet**

***Index Terms*—Frequency-domain learning, hyperspectral image classification, learnable spectral filtering, spectral–spatial interaction.**

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## I. Introduction

MULTIBAND visual sensing has become an important paradigm for fine-grained scene understanding, as it records dense spectral responses together with spatial structures. Compared with conventional RGB imagery, hyperspectral images provide hundreds of contiguous narrow bands, making it possible to distinguish visually similar materials according to subtle reflectance variations, absorption characteristics, and spectral continuity. These properties have made hyperspectral image classification a fundamental tool for land-cover analysis and agricultural monitoring [1], as well as material-aware interpretation in resource exploration and fine-grained remote sensing applications [2]. From a visual-system perspective, hyperspectral classification can be viewed as a representative multiband image understanding problem, where high-dimensional spectral signals and spatial structures need to be jointly represented in an accurate, compact, and efficient manner. Recent studies on cross-scene and heterogeneous hyperspectral recognition further indicate that such models should not be evaluated only under closed-set benchmark conditions [3], [4].

Despite considerable progress, effective hyperspectral representation learning remains nontrivial. Spectral bands are not isolated channels; they form continuous responses shaped by material composition, imaging conditions, and sensor characteristics. The discriminative cues are therefore often embedded in spectral trends, local oscillations, and energy variations across bands rather than in individual band responses. Meanwhile, hyperspectral scenes contain heterogeneous spatial patterns, including object boundaries, mixed pixels, and fine-grained local textures. These factors require a model to propagate spectral cues into neighborhood-level spatial structures while avoiding high computational cost, memory usage, and inference latency.

Existing deep learning methods have approached this problem from different perspectives. CNN-based models exploit local spatial inductive biases and have achieved promising performance through 2-D, 3-D, and hybrid convolutional structures [5], [6]. However, fixed convolutional kernels have limited ability to characterize long-range spectral dependencies and global structural variations. Transformer-based models improve global interaction through self-attention [7], [8], but their quadratic token complexity introduces substantial

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computational and memory overhead for high-dimensional hyperspectral data. Domain-adaptive and cross-scene methods enhance transferability under distribution shifts [3], [9], yet they do not directly address the transform-domain structure of spectral responses. Recently, state-space models have been introduced into hyperspectral image classification because of their linear-complexity sequence modeling ability [10], [11]. These methods are effective in long-range dependency modeling, but most of them still organize hyperspectral data as spatial–spectral sequences in the original feature domain. Consequently, the explicit modeling of frequency-domain spectral variations and their interaction with spatial structures remains insufficient.

Frequency-domain learning has been explored in multiple ways for hyperspectral and visual representation learning. Some methods use Fourier-, cosine-, wavelet-, or other transforms to obtain transform-domain representations, while others incorporate transform-domain priors as auxiliary network modules [12], [13], [14], [15]. These strategies are valuable, but the key distinction in our work is not between frequency-domain learning and nonfrequency learning per se; rather, it concerns how the frequency response is used. In the spectral–spatial learning adaptive network (SSLANet), the frequency response is learned end-to-end and conditioned on the current input feature distribution, instead of being treated as a fixed representation or a static preprocessing step. For hyperspectral classification, this distinction matters because the informative spectral components vary with material category, local context, and imaging condition. Therefore, the effective frequency transfer function should be modulated according to the current frequency energy and remain differentiable during end-to-end optimization.

The frequency-domain view in SSLANet is not intended as an intrinsically superior replacement for the spatial–spectral representation. Since the Fourier transform is invertible, the transform itself does not create new discriminative information. Instead, it provides a complementary basis in which the learned latent representation can be selectively modulated according to the current input through normalized frequency energy, an adaptive soft mask, and the effective transfer function $H(F)$. The resulting frequency-domain operation therefore complements the local spatial interaction modeled by the interactive convolution block (ICB) rather than replacing it.

Motivated by this observation, we propose SSLANet, a compact frequency–spatial learning network for efficient hyperspectral image classification. Instead of treating spectral bands merely as channel responses in the original feature space, SSLANet maps latent spectral–channel features into the frequency domain and performs learnable complex-valued filtering. Its core unit is a residual frequency–spatial learning layer composed of an adaptive spectral block (ASB) and an ICB. ASB conducts complex-valued spectral modulation and current-input-dependent frequency-energy selection, producing an input-conditioned frequency transfer function that adaptively emphasizes discriminative spectral components. ICB then propagates the filtered responses into local spatial neighborhoods through multiplicative spatial–channel interaction, which provides a compact alternative to dense high-order feature coupling. By integrating these two operators with residual updates, SSLANet jointly strengthens frequency-domain spectral modeling and local spatial structure perception while maintaining efficient inference.

The main contributions of this work are summarized as follows.

1) We model hyperspectral spectral representation learning as an input-conditioned frequency-domain filtering process. The proposed ASB maps latent spectral–channel features into frequency responses and performs learnable complex-valued modulation together with current-input-dependent normalized frequency-energy selection.
2) We introduce an energy-adaptive spectral selection mechanism based on a soft frequency mask $M(F)$. With a learnable threshold and temperature-controlled sigmoid selection, the proposed design continuously optimizes discriminative frequency components without relying on nondifferentiable hard filtering.
3) We design an ICB to model spatial propagation through factorized multiplicative spatial–channel interaction. The multiplicative coupling between pointwise channel recomposition and local neighborhood aggregation captures second-order interaction cues without explicitly constructing dense bilinear tensors.
4) We build a residual frequency–spatial learning framework and evaluate it on three public hyperspectral datasets. Extensive experiments demonstrate that SSLANet achieves a favorable balance between classification performance and computational cost.

## II. Related Works

### A. Spectral–Spatial Learning for Multiband Remote Sensing Vision

Hyperspectral image classification is a representative multiband visual understanding task, where each pixel is associated with a dense spectral signature and a spatial neighborhood. Before deep learning became dominant, support vector machines and discriminant analysis were widely used for spectral recognition [16], [17]. Dimensionality-reduction and locality-preserving strategies further improved feature compactness under high spectral redundancy [18], [19]. Early deep learning methods mainly relied on CNNs to extract local spectral–spatial features. 1-D CNNs emphasize spectral responses, 2-D CNNs focus on local spatial structures, and 3-D CNNs jointly process spectral and spatial dimensions. Hybrid convolutional architectures then combine 3-D and 2-D operators to balance representation ability and computational cost [6], [20]. More recent heterogeneous and cross-scene designs suggest that spectral–spatial learning should also account for distribution shifts across visual scenarios [4], [9]. Although convolutional models provide strong local inductive biases, their receptive fields are inherently constrained by kernel size, making it difficult to capture long-range band dependencies and global scene structures.

Transformer-based models have been introduced to overcome the locality limitation of CNNs. By tokenizing hyperspectral patches and applying self-attention, these

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methods can capture global spectral–spatial dependencies. SpectralFormer and HiT demonstrate the effectiveness of global token interaction, while DCTN, SS-TMNet, and recent attention variants further enhance hierarchical and adaptive spectral–spatial representations [7], [8], [21], [22], [23], [24]. However, the quadratic complexity of self-attention results in considerable computational and memory costs when processing large spatial patches or dense spectral sequences and may also increase the risk of overfitting under limited supervision.

Recently, state-space models and Mamba-based architectures have attracted attention because of their linear-complexity sequence modeling ability. Representative models such as 3DSS-Mamba and $S^2$Mamba show the potential of selective scanning for hierarchical spectral–spatial representation [10], [11], [25]. MambaHSI and related variants further adapt state-space modeling to hyperspectral tokens and spatial contexts [26], [27]. In addition, WaveMamba uses wavelet decomposition and selective scanning to jointly model low-frequency and high-frequency features [28]. Semi-Mamba and RSRWKV indicate that linear-complexity context aggregation is promising for efficient visual systems [29], [30]. Although transform-aware methods such as WaveMamba incorporate frequency-domain information, most Mamba-based models still perform sequence scanning primarily in the original feature domain. In contrast, SSLANet combines input-conditioned latent frequency-domain spectral filtering with local spatial interaction, providing a complementary design to CNN-, Transformer-, and Mamba-based paradigms.

Recent few-shot hyperspectral classification studies further address the limited-annotation setting. Yan et al. [31] propose semantic-anchor-enhanced prototype learning, where trainable semantic anchors are used to improve class-prototype representation under scarce supervision. For the more challenging cross-domain few-shot setting, Zhang et al. [32] develop orthogonal feature disentanglement to separate domain-shared and domain-specific subspaces, reducing domain-specific interference while preserving transferable discriminative information. These studies highlight the growing importance of label efficiency and cross-domain generalization in hyperspectral recognition.

### *B. Frequency-Domain Visual Learning and Transform-Based Representation*

Frequency-domain analysis has long been a fundamental tool in image and video processing, including filtering, compression, denoising, restoration, and representation learning. Compared with spatial-domain operations, transform-domain methods can reveal global signal structures, energy distributions, and periodic variations that are not easily captured by local kernels. In multiband visual data, spectral responses across contiguous bands naturally form structured signals. Discriminative material information may appear as spectral trends, local oscillations, absorption-related variations, or high-energy frequency components.

Existing frequency-domain learning methods usually employ Fourier, cosine, or wavelet transforms to improve robustness and representation capacity. Fractional Fourier and multilevel frequency modeling have been explored for hyperspectral representation learning [12], [13]. Frequency-oriented visual Transformers further demonstrate that transform-domain priors can reduce redundant token interactions in efficient vision models [14]. For restoration-oriented hyperspectral systems, spatial–spectral regularization in the transform or recovery domain is also beneficial for preserving structural consistency [33]. Our work is related to this family of methods, but it differs in the way the frequency response is used: the proposed ASB is not a fixed transform or a static band-selection operator. Instead, it integrates learnable complex-valued modulation with the current normalized frequency energy to generate a threshold- and temperature-controlled soft mask $M(F)$ and an effective input-conditioned transfer function $H(F)$. ASB therefore performs differentiable and adaptive frequency filtering according to the current input feature distribution, selectively emphasizing discriminative frequency components throughout end-to-end optimization.

Unlike transform operations applied directly to raw contiguous spectral bands, ASB performs the Fourier transform along the learned latent spectral–channel dimension after embedding. Consequently, the resulting latent frequency indices describe organization in the learned feature basis and should not be interpreted as having a one-to-one physical correspondence with the original hyperspectral wavelengths.

### *C. Efficient Visual Systems for Remote Sensing Image Understanding*

For remote sensing and multiband image understanding, system efficiency is as important as recognition accuracy. Practical visual systems are often constrained by GPU memory, inference latency, limited annotation, and large-scale scene coverage. These constraints are especially evident in high-resolution agricultural scenes and UAV-borne hyperspectral applications [34], [35]. Recent efficient remote sensing systems have therefore emphasized lightweight context modeling and scalable multimodal fusion [30], [36]. Semisupervised Mamba-style learning also reflects the need to reduce annotation and computation burdens simultaneously [29]. CNNs are efficient but limited in global modeling; Transformers are expressive but computationally expensive; state-space models improve sequence efficiency but may not fully exploit frequency-domain priors. Therefore, an effective multiband visual system should simultaneously consider representation ability, computational complexity, and implementation cost.

Related transform-aware learning has also been developed for adjacent multimodal remote-sensing problems. CIRSM-Net studies cyclic SAR–optical registration, while DADIGAN uses disentangled iterative generation with attention for SAR/optical cloud-and-shadow removal [37], [38]. Because these methods address tasks different from hyperspectral image classification, we cite them as multimodal remote sensing context rather than as directly comparable classification baselines.

Following this principle, SSLANet integrates FFT-based latent frequency-domain filtering with factorized multiplicative spatial–channel interaction in a residual architecture, without explicitly constructing a dense second-order interaction tensor.

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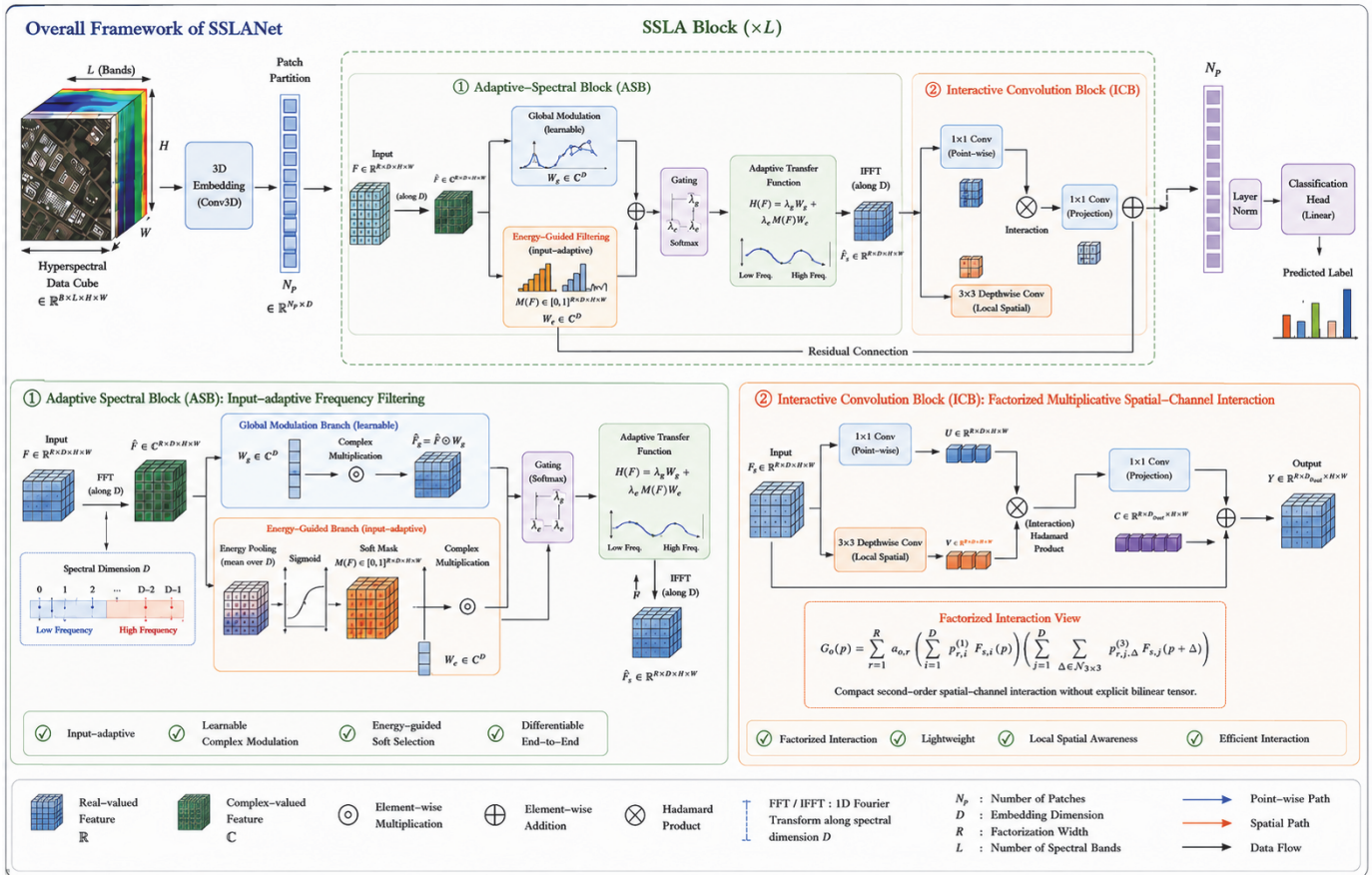



Fig. 1. Overall architecture of the proposed SSLANet for efficient multiband visual understanding. The network first embeds hyperspectral patches into a compact feature space, then stacks frequency–spatial learning layers composed of the ASB and ICB. ASB performs learnable spectral filtering in the frequency domain, while ICB models factorized multiplicative spatial–channel interaction. The final prediction is obtained through dual-branch feature aggregation and classification.

Rather than claiming universal superiority across all efficiency metrics, it aims to provide a practical accuracy–efficiency tradeoff for hyperspectral image classification.

## III. Proposed Methods

### A. Overall Architecture

Let the input hyperspectral patch be denoted as

$$X \in \mathbb{R}^{B\times L\times P\times P} \tag{1}$$

where $B$ is the batch size, $L$ is the number of spectral bands, and $P$ is the spatial patch size. SSLANet first maps the input patch into a compact latent feature tensor

$$F_0 = f_{\text{emb}}(X), \quad F_0 \in \mathbb{R}^{B\times D\times H\times W} \tag{2}$$

where $D$ denotes the embedding dimension, and $H \times W$ denotes the spatial size after feature embedding.

The embedded feature is then processed by $N$ stacked frequency–spatial learning layers. For the $n$th layer, the update rule is

$$F'_n = F_{n-1} + \text{ASB}(\text{Norm}_1(F_{n-1})) \tag{3}$$

$$F_n = F'_n + \text{ICB}(\text{Norm}_2(F'_n)). \tag{4}$$

This design follows the residual structure in Fig. 1: ASB first enhances latent spectral–channel responses in the frequency domain, and ICB subsequently propagates the enhanced information into local spatial neighborhoods.

After the stacked learning layers, the final feature $F_N$ is aggregated through global average pooling and global max pooling. The concatenated representation is then fed into the classification head.

$$z_{\text{avg}} = \text{Flatten}(\text{GAP}(F_N)), \quad z_{\max} = \text{Flatten}(\text{GMP}(F_N))$$
$$\hat{y} = f_{\text{cls}}(\text{Concat}[z_{\text{avg}}, z_{\max}]) \tag{5}$$

where GAP(·) and GMP(·) denote global average pooling and global max pooling, respectively. The classification head consists of a linear layer, dropout, and a second linear layer, producing raw logits. When class probabilities are required, softmax is applied to the logits.

### B. Spatial Feature Embedding

The spatial feature embedding module is used to convert the original hyperspectral patch into a latent spectral–spatial representation. Different from purely 2-D spatial projection, the embedding stem adopts 3-D convolution to jointly capture local spectral variations and spatial context within the input patch

$$F_0 = f_{\text{emb}}(X) = \phi(\text{Norm}(\text{Conv3D}(X))) \tag{6}$$

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where Conv3D(·) denotes the 3-D convolutional embedding operation, Norm(·) is a normalization layer, and $\phi(\cdot)$ is the activation function.

The output of this module is reshaped or projected into

$$F_0 \in \mathbb{R}^{B\times D\times H\times W}. \tag{7}$$

Here, $D$ denotes the learned embedding dimension rather than the number of original spectral bands. The 3-D convolution first operates on contiguous input bands to capture local spectral adjacency, after which reshaping and a learned $1 \times 1$ projection mix the intermediate spectral–spatial responses. Consequently, the final latent channels no longer maintain a strict one-to-one or monotonic correspondence with the original wavelength axis but instead form a learned latent feature basis. The frequency-domain operation in ASB therefore acts on this latent spectral–channel representation rather than on the raw physical wavelength sequence.

### C. Frequency–Spatial Learning Layer

Each frequency–spatial learning layer consists of two sequential residual units: ASB and ICB. Given the input feature $F_{n-1}$, the layer first normalizes it and applies ASB

$$\widetilde{F}_{n-1} = \mathrm{Norm}_1\left(F_{n-1}\right), \quad S_n = \mathrm{ASB}\left(\widetilde{F}_{n-1}\right). \tag{8}$$

The ASB output is added back to the input feature.

$$F'_n = F_{n-1} + S_n. \tag{9}$$

Then, the intermediate feature is normalized and passed into ICB.

$$\widetilde{F}'_n = \mathrm{Norm}_2\left(F'_n\right), \quad G_n = \mathrm{ICB}\left(\widetilde{F}'_n\right). \tag{10}$$

The final output of the layer is

$$F_n = F'_n + G_n. \tag{11}$$

This residual update preserves the original representation while progressively injecting frequency-enhanced spectral information and spatial–channel interaction cues.

*1) Adaptive Spectral Block:* The ASB is designed as a learnable spectral filtering operator in the latent frequency domain. Given an input feature

$$F \in \mathbb{R}^{B\times D\times H\times W}. \tag{12}$$

ASB first performs a 1-D real Fourier transform along the latent spectral–channel dimension. Since the input features are real-valued, only the nonredundant frequency components are retained.

$$\widehat{F} = \mathcal{F}_D^{\mathrm{r}}(F), \qquad \widehat{F} \in \mathbb{C}^{B\times D_f\times H\times W}, \quad D_f = \left\lfloor \frac{D}{2} \right\rfloor + 1 \tag{13}$$

where $\mathcal{F}_D^{\mathrm{r}}(\cdot)$ denotes the 1-D real FFT along dimension $D$. For the adopted embedding dimension $D = 256$, this produces $D_f = 129$ nonredundant latent frequency components. Because the learned $1 \times 1$ projection mixes intermediate spectral–spatial responses, $\widehat{F}$ represents the frequency response of the latent spectral–channel features rather than the physical spectrum of the original reflectance sequence. Therefore, the latent frequency indices have no one-to-one correspondence with the original wavelengths.

For a latent vector $\mathbf{x} = [x_0, \ldots, x_{D-1}]^T$ at a fixed spatial location, its $k$th Fourier coefficient is

$$\widehat{x}_k = \sum_{d=0}^{D-1} x_d \exp\left(-j\frac{2\pi kd}{D}\right). \tag{14}$$

For real-valued $\mathbf{x}$, the real FFT explicitly retains only the nonredundant coefficients $k = 0, \ldots, D_f - 1$, while the remainder are determined by conjugate symmetry. Each coefficient is a global linear combination of responses over the latent channel dimension. Since the Fourier transform is invertible, it provides an alternative representation basis without creating new discriminative information. The useful operation in ASB is the subsequent input-conditioned modulation $E(F) \to M(F) \to H(F)$, which selectively reweights the latent representation according to the current feature statistics.

As illustrated in Fig. 1, ASB contains two complementary branches. The first branch performs global spectral modulation

$$\widehat{F}_g = \widehat{F} \odot W_g \tag{15}$$

where $W_g \in \mathbb{C}^{D_f}$ is a learnable complex-valued filter and $\odot$ denotes elementwise multiplication. The complex filter is parameterized by real and imaginary parts

$$W_g = W_g^{\mathrm{re}} + iW_g^{\mathrm{im}}, \quad W_g^{\mathrm{re}}, W_g^{\mathrm{im}} \in \mathbb{R}^{D_f}. \tag{16}$$

The second branch performs energy-guided selective filtering. The normalized latent frequency energy is computed as

$$E_{b,d,h,w} = \frac{\left|\widehat{F}_{b,d,h,w}\right|^2}{\epsilon + \frac{1}{D_f}\sum_{d'=1}^{D_f}\left|\widehat{F}_{b,d',h,w}\right|^2} \tag{17}$$

where the mean operation is taken over the $D_f$ nonredundant latent frequency components for each sample and spatial position. The denominator is broadcast to all retained frequency indices.

A differentiable soft energy mask is then generated by

$$M_{b,d,h,w} = \sigma\left(\frac{E_{b,d,h,w} - \tau_d}{T}\right), \quad T > 0 \tag{18}$$

where $\tau_d$ is a learnable frequency-dependent threshold and $T$ is a positive temperature parameter. The temperature is not a fixed constant. In the default configuration, all ASB layers are initialized with $T_0 = 1.0$, after which each layer learns its own temperature independently during training. To ensure positivity, $T$ is parameterized as

$$T = \mathrm{softplus}\left(\theta_T\right) + \epsilon_T. \tag{19}$$

This parameter controls the sharpness of the soft mask. As $T \to 0^+$, the sigmoid approaches a hard threshold, yielding sharper frequency selection. As $T$ becomes large, $(E_{b,d,h,w} - \tau_d)/T \to 0$ and $M_{b,d,h,w} \to 0.5$, flattening the mask and weakening its selectivity. These limiting behaviors characterize the mask itself and do not imply a monotonic accuracy trend. The energy-guided branch is written as

$$\widehat{F}_e = \left(\widehat{F} \odot M\right) \odot W_e \tag{20}$$

where $W_e \in \mathbb{C}^{D_f}$ is another learnable complex-valued filter

$$W_e = W_e^{\mathrm{re}} + iW_e^{\mathrm{im}}, \quad W_e^{\mathrm{re}}, W_e^{\mathrm{im}} \in \mathbb{R}^{D_f}. \tag{21}$$

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The two branches are adaptively fused through a softmax gating mechanism.

$$\left[\lambda_g, \lambda_e\right] = \text{Softmax}\left(\left[a_g, a_e\right]\right) \tag{22}$$

where $a_g$ and $a_e$ are learnable scalar parameters. Beyond viewing the two outputs as separate branches, ASB can be equivalently interpreted as an input-conditioned transfer function in the latent frequency domain. Substituting (15) and (20) into the fusion rule yields

$$\widehat{F}_s = \lambda_g \widehat{F}_g + \lambda_e \widehat{F}_e = \widehat{F} \odot H(F) \tag{23}$$

where the effective frequency transfer function is defined as

$$H_{b,d,h,w}(F) = \lambda_g W_{g,d} + \lambda_e M_{b,d,h,w}(F) W_{e,d}. \tag{24}$$

Here, $H(F)$ is not a fixed transform-domain filter. The first term, $\lambda_g W_{g,d}$, provides a globally shared spectral–channel response learned from the training data, while the second term, $\lambda_e M_{b,d,h,w}(F) W_{e,d}$, introduces sample- and location-dependent frequency selection through the current-input-dependent normalized frequency energy. Therefore, ASB implements an input-conditioned spectral transformation rather than a fixed transform preprocessing.

The filtering behavior of (24) is controlled by three interpretable factors. The complex filters $W_g$ and $W_e$ modulate both amplitude- and phase-related latent frequency responses; the learnable threshold $\tau_d$ determines which frequency components are emphasized relative to their normalized energy; and the temperature $T$ controls the sharpness of the transition between suppressed and enhanced components. When $E_{b,d,h,w}$ is lower than $\tau_d$, the mask response tends to be small, and the selective branch is weakened. When $E_{b,d,h,w}$ exceeds $\tau_d$, the corresponding component is smoothly preserved and amplified by the selective filter. Compared with hard frequency truncation, this soft transfer function remains differentiable and can be optimized end-to-end.

Finally, the filtered representation is transformed back to the real feature domain.

$$F_s = \left(\mathcal{F}_D^{\mathrm{r}}\right)^{-1}\left(\widehat{F} \odot H(F)\right). \tag{25}$$

The inverse real FFT reconstructs the real-valued latent feature representation with the original channel dimension $D$. Under an orthonormal Fourier basis, the feature energy after ASB is governed by the magnitude of the effective transfer function

$$\|F_s\|_2 \le \Gamma(F)\|F\|_2, \quad \Gamma(F) = \max_{b,d,h,w}\left|H_{b,d,h,w}(F)\right|. \tag{26}$$

Thus, the spectral response of ASB can be interpreted as a learnable gain control mechanism. The surrounding normalization and residual update further prevent the frequency-filtered response from replacing the original representation abruptly, which improves optimization stability in small-sample hyperspectral classification.

*2) Interactive Convolution Block:* The ICB propagates the ASB-enhanced responses into spatial neighborhoods. Given the input feature $F_s \in \mathbb{R}^{B\times D\times H\times W}$, ICB uses two parallel convolutional branches

$$U = \phi(P_1 * F_s) \tag{27}$$

$$V = \phi(P_3 * F_s) \tag{28}$$

where $P_1$ denotes a $1 \times 1$ pointwise convolution, $P_3$ denotes a $3 \times 3$ local convolution, and $\phi(\cdot)$ is the activation function. The $1 \times 1$ branch recomposes latent channels, while the $3 \times 3$ branch aggregates local spatial context.

The two branches interact through elementwise multiplication.

$$Q = U \odot V. \tag{29}$$

The interacted feature is then projected by a $1 \times 1$ convolution

$$G = P_o * Q \tag{30}$$

where $P_o$ denotes the output projection.

Because the two branches in (27) and (28) include the nonlinear activation $\phi(\cdot)$, the implemented ICB is not, in general, a bilinear map of $F_s$. We therefore use factorization only to describe its structural parameterization, rather than to claim a strict low-rank approximation. Let the preactivation branch responses be $\widetilde{U} = P_1 * F_s$ and $\widetilde{V} = P_3 * F_s$. For a fixed output channel $o$, their multiplicative interaction followed by $P_o$ can be expanded as:

$$\widetilde{G}_o(p) = \sum_{r=1}^{R} a_{o,r}\left(\sum_{i=1}^{D} p_{r,i}^{(1)} F_{s,i}(p)\right) \times \left(\sum_{j=1}^{D}\sum_{\Delta\in\mathcal{N}_{3\times 3}} p_{r,j,\Delta}^{(3)} F_{s,j}(p+\Delta)\right) \tag{31}$$

where $R$ is the interaction width, $p_{r,i}^{(1)}$ and $p_{r,j,\Delta}^{(3)}$ are the branch coefficients, and $a_{o,r}$ is the output-projection coefficient. In this preactivation structural case, the induced second-order kernel is

$$\mathcal{K}_{o,i,j,\Delta} = \sum_{r=1}^{R} a_{o,r} p_{r,i}^{(1)} p_{r,j,\Delta}^{(3)}. \tag{32}$$

Equation (32) is a sum of $R$ separable factors. Thus, ICB is best interpreted as a factorized multiplicative spatial–channel interaction. Since the reported setting uses $R = D = 256$, no strict low-rank bottleneck is imposed. The practical benefit is that the dense second-order interaction tensor is never explicitly constructed.

*3) Complexity Analysis of ASB and ICB:* Given an input feature $F \in \mathbb{R}^{B\times D\times H\times W}$, ASB performs 1-D FFT and an inverse FFT along the latent spectral–channel dimension $D$. The main computational cost is therefore,

$$\mathcal{O}_{\text{ASB}} = \mathcal{O}\left(BHWD\log_2 D\right) \tag{33}$$

while the energy normalization, soft mask generation, complex modulation, and branch fusion only introduce linear elementwise operations of $\mathcal{O}(BHWD)$. Thus, ASB scales quasi-linearly with the latent spectral dimension. In contrast, spectral self-attention over $D$ latent tokens requires $\mathcal{O}(BHWD^2)$ attention computation and $\mathcal{O}(BHWD^2)$ attention storage. This gives ASB a clearer efficiency advantage when $D$ increases.

For ICB, let $R$ denote the interaction width or factorization width of the two convolutional branches. The $1 \times 1$ branch,

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$3 \times 3$ branch, and output projection are configured as $D \to R$, $D \to R$, and $R \to D$, respectively. Ignoring bias terms, the parameter number is

$$\theta_{\mathrm{ICB}} = DR + 9DR + RD = 11DR. \tag{34}$$

The corresponding computational cost is

$$\mathcal{O}_{\mathrm{ICB}} = \mathcal{O}\left(BHW\left(DR + 9DR + RD\right)\right) = \mathcal{O}\left(11BHWDR\right). \tag{35}$$

In all reported experiments, we set $R = D$, and the embedding dimension is $D = 256$. Therefore, the actual ICB complexity becomes

$$\mathcal{O}_{\mathrm{ICB}} = \mathcal{O}\left(11BHWD^2\right) \tag{36}$$

which is consistent with the implemented parameter count $\theta_{\mathrm{ICB}} = 11D^2$. This efficiency advantage comes from using a factorized parameterization instead of explicitly constructing a dense second-order spatial–channel interaction tensor. A bottleneck setting with $R < D$ would further reduce the computational cost and parameter count, but it is not used in the reported experiments.

### D. Dual-Branch Feature Aggregation and Classification

After the stacked frequency–spatial learning layers, SSLANet obtains the final feature tensor.

$$F_N \in \mathbb{R}^{B\times D\times H\times W}. \tag{37}$$

As shown in Fig. 1, dual-branch aggregation summarizes the final representation with two complementary statistics. Global average pooling captures the mean contextual response across the patch, whereas global max pooling retains the strongest channelwise activations that can be attenuated by averaging. Their concatenation therefore preserves both distributed context and salient responses before classification, without assuming that either statistic is universally superior.

$$z_{\mathrm{avg}} = \mathrm{Flatten}\left(\mathrm{GAP}\left(F_N\right)\right) \tag{38}$$

$$z_{\max} = \mathrm{Flatten}\left(\mathrm{GMP}\left(F_N\right)\right). \tag{39}$$

The final representation is obtained by concatenation.

$$z = \mathrm{Concat}\left[z_{\mathrm{avg}}, z_{\max}\right]. \tag{40}$$

For inference, the class probabilities are obtained by applying softmax to the raw logits produced by the classification head.

$$\hat{y} = \mathrm{Softmax}\left(W_2\mathrm{Dropout}\left(\phi\left(W_1 z\right)\right)\right) \tag{41}$$

where $W_1$ and $W_2$ denote the two linear layers. During training, the raw logits before softmax are directly used to compute the masked cross-entropy loss.

## IV. Experiments

### A. Datasets Description

To comprehensively evaluate the effectiveness and generalization capability of the proposed SSLANet, experiments were conducted on three representative public hyperspectral datasets: Indian Pines [39], Houston 2013 [40], and WHU-Hi-LongKou [34].

*1) Indian Pines:* This dataset was acquired by the AVIRIS sensor, covering agricultural areas in northwestern Indiana, USA. The raw data comprise 224 spectral bands spanning wavelengths from 0.4 to 2.5 $\mu$m. After removing water absorption bands, 200 effective bands remain. The spatial dimensions are $145 \times 145$ pixels, with a spatial resolution of approximately 20 m. The dataset contains 10 249 labeled pixels representing 16 land cover classes. Due to subtle spectral differences between classes and class imbalance, this dataset is widely used to evaluate the performance and robustness of hyperspectral classification algorithms. This study employs 10% of the labeled samples as the training set, with the remaining samples serving as the test set.

*2) Houston 2013:* The Houston 2013 dataset was acquired by an airborne hyperspectral sensor over the urban area of Houston, Texas, USA. The data comprise 144 effective spectral bands with an image resolution of $1905 \times 349$ pixels. The dataset includes 15 urban land cover classes, with a total of 15 029 valid annotated samples. This study employs a $15 \times 15$ sliding window to extract combined spatial–spectral features. The dataset is divided into training and test sets based on a 1% labeling ratio, comprising 143 training samples and 14 886 test samples.

*3) WHU-Hi-LongKou:* This dataset, released by Wuhan University, is a typical high-spatial-resolution hyperspectral dataset for agricultural scenarios. It contains 270 effective spectral bands with an image size of $550 \times 400$ pixels. The dataset includes 9 land cover categories, with a total of 204 542 valid annotated samples. This study employs a $15 \times 15$ sliding window to extract combined spatial–spectral features. The dataset is divided into training and test sets based on a 0.1% annotation ratio, comprising 202 training samples and 204 340 test samples. It should be noted that the reported results follow the fixed random split protocol in Table I.

### B. Experimental Setup

*1) Implementation Details:* All methods are evaluated under the same data preprocessing and training configuration. The hyperspectral data are normalized before being fed into the network.

SSLANet is constructed with four frequency–spatial learning layers. The embedding dimension is set to 256, and the dropout rate is set to 0.1. The model is trained using the AdamW optimizer [41], with an initial learning rate of $10^{-3}$ and a weight decay of $10^{-4}$. A cosine annealing strategy [42] is adopted to adjust the learning rate during training. The maximum number of training epochs is 200, and early stopping with a patience of 20 epochs is used to reduce overfitting. All experiments are implemented in PyTorch and conducted on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory.

We employ a masked cross-entropy loss to optimize SSLANet. The training objective is

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{\sum_{i=1}^{B} w_{y_i}} \sum_{i=1}^{B} w_{y_i} \log \frac{\exp\left(z_{i,y_i}\right)}{\sum_{c=1}^{C} \exp\left(z_{i,c}\right)} \tag{42}$$

where $B$ and $C$ denote the batch size and the number of classes, respectively; $z_{i,c}$ is the logit of sample $i$ for class $c$,

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TABLE I
TRAINING AND TEST SAMPLE NUMBERS IN THE INDIAN PINES, HOUSTON 2013 AND WHU-HI-LONGKOU DATASETS

| | Indian Pines (10% labeled) | | Houston 2013 (1% labeled) | | WHU-Hi-LongKou (0.1% labeled) | |
|---|---|---|---|---|---|---|
| No. | Class Name | Train / Test | Class Name | Train / Test | Class Name | Train / Test |
| 1 | Alfalfa | 5 / 41 | Healthy grass | 12 / 1239 | Corn | 35 / 34476 |
| 2 | Corn-notill | 142 / 1286 | Stressed grass | 12 / 1242 | Cotton | 8 / 8366 |
| 3 | Corn-mintill | 83 / 747 | Synthetic grass | 6 / 691 | Sesame | 3 / 3028 |
| 4 | Corn | 24 / 213 | Trees | 12 / 1232 | Broad-leaf soybean | 63 / 63149 |
| 5 | Grass-pasture | 48 / 435 | Soil | 12 / 1230 | Narrow-leaf soybean | 4 / 4147 |
| 6 | Grass-trees | 73 / 657 | Water | 3 / 322 | Rice | 12 / 11842 |
| 7 | Grass-pasture-mowed | 3 / 25 | Residential | 12 / 1256 | Water | 67 / 66989 |
| 8 | Hay-windrowed | 48 / 430 | Commercial | 12 / 1232 | Roads and houses | 7 / 7117 |
| 9 | Oats | 2 / 18 | Road | 12 / 1240 | Mixed weed | 6 / 5223 |
| 10 | Soybean-notill | 97 / 875 | Highway | 12 / 1215 | | |
| 11 | Soybean-mintill | 246 / 2209 | Railway | 12 / 1223 | | |
| 12 | Soybean-clean | 59 / 534 | Parking lot 1 | 12 / 1221 | | |
| 13 | Wheat | 21 / 184 | Parking lot 2 | 4 / 465 | | |
| 14 | Woods | 126 / 1139 | Tennis court | 4 / 424 | | |
| 15 | Buildings | 38 / 348 | Running track | 6 / 654 | | |
| 16 | Stone | 9 / 84 | | | | |
| | **Total: 1024 / 9225** | | **Total: 143 / 14886** | | **Total: 202 / 204340** | |

$y_i$ is the ground-truth label, and $w_{y_i}$ is the corresponding class weight. The weight is set to 1 for valid classes and 0 for ignored labels; when all samples in a batch belong to valid classes, (42) reduces to the standard mean cross-entropy loss. No label smoothing, auxiliary loss, or additional task-specific loss is used. Training operates directly on the raw logits, while softmax is applied only to obtain class probabilities during inference. The AdamW weight decay of $1 \times 10^{-4}$ is optimizer-level regularization and is not included in the training objective.

For the adopted $15 \times 15$ spatial window, no zero, reflection, or other artificial padding is applied to the original hyperspectral scene. Training and validation use valid-center patch extraction, in which only pixels whose complete $15 \times 15$ neighborhoods lie inside the image are selected as centers. Consequently, the outer seven-pixel margin is excluded as training or validation centers. During full-scene inference, sliding windows are likewise constrained to remain inside the image and are shifted inward near the boundaries when necessary. Predictions from valid overlapping windows are accumulated over their covered locations, allowing border pixels to be classified without introducing synthetic spectral–spatial values.

For efficiency evaluation, parameters, FLOPs, FPS, and training time are reported under a unified benchmark configuration. Throughput on Indian Pines is measured on an NVIDIA GeForce RTX 4090 GPU using $15 \times 15$ input patches, a batch size of 100, and FP32 inference without mixed precision. Six minibatches are first processed for GPU warm-up, followed by CUDA synchronization immediately before and after one complete pass through the test loader. FPS is computed as the total number of processed samples divided by the total wall-clock time, including DataLoader iteration and host-to-device transfer.

*2) Evaluation Metrics:* We use overall accuracy (OA), average accuracy (AA), and the Kappa coefficient as evaluation metrics [43]. OA measures the proportion of correctly classified test samples over all test samples. AA computes the mean accuracy over all classes and is therefore more sensitive to class imbalance. The Kappa coefficient evaluates the agreement between predictions and ground-truth labels while considering chance agreement. In addition, classwise accuracy is reported to provide a more detailed analysis of category-level discrimination.

### C. Competitive Methods

To evaluate the proposed frequency–spatial learning design, SSLANet is compared with representative methods from convolutional, Transformer-based, and state-space model families. The convolutional baselines include ***2-D CNN*** and ***HybridSN***. The 2-D CNN mainly extracts local spatial patterns [5], [44]. HybridSN combines 3-D and 2-D convolutions to balance spectral–spatial representation and computational cost [6].

For Transformer-based comparison, we include ***DCTN***, ***HiT***, ***SS-TMNet***, ***SSFTTNet***, and ***SpectralFormer***. These methods introduce token interaction, hierarchical Transformer encoding, or spatial–spectral attention to capture global dependencies [7], [8], [21], [22], [45]. For state-space-based comparison, we include ***MambaHSI***, ***3DSS-Mamba***, ***S$^2$Mamba***, and ***WaveMamba*** [10], [11], [26], [28]. These methods provide linear-complexity sequence modeling and are closely related to recent efficient hyperspectral representation learning. Together, these baselines cover local convolution, global attention, and state-space sequence modeling, providing a broad reference for evaluating SSLANet.

### D. Experimental Comparison With Competitive Approaches

This article systematically compares SSLANet with current state-of-the-art methods on three public hyperspectral datasets (Indian Pines, Houston 2013, and WHU-Hi-LongKou). The evaluation metrics include OA (%), AA (%), and Kappa (%), which are summarized in Tables II–IV, and the corresponding visualizations are shown in Figs. 2–4.

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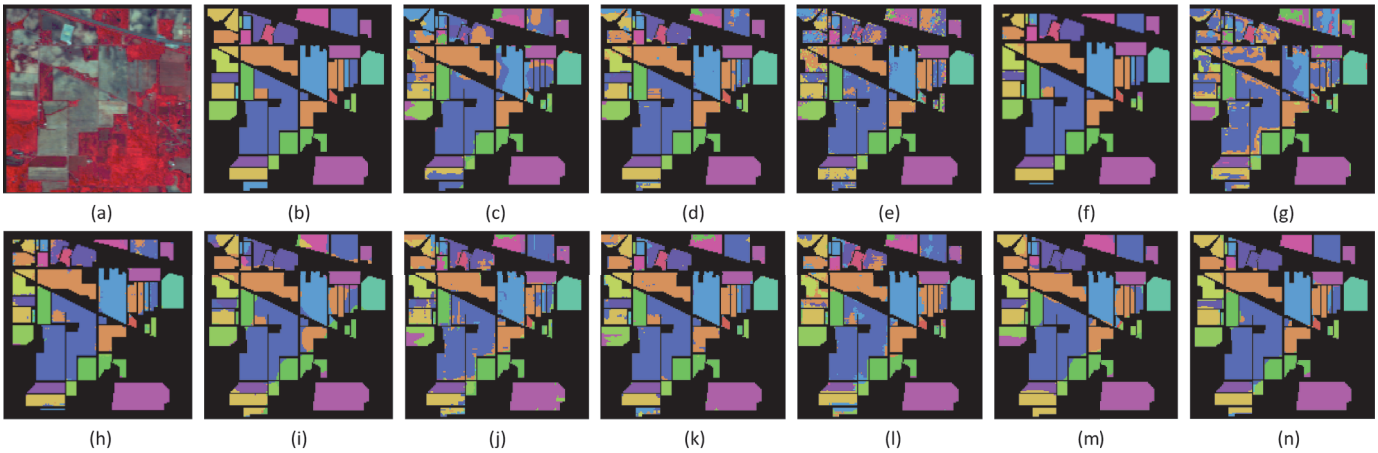



Fig. 2. Visual comparison of the Indian pines dataset and classification maps obtained by different methods. (a) RGB composite. (b) Ground truth. (c) 2-D-CNN. (d) HybridSN. (e) DCTN. (f) HiT. (g) SS-TMNet. (h) SSFTTNet. (i) SpectralFormer. (j) MambaHSI. (k) 3DSS-Mamba. (l) $S^2$Mamba. (m) WaveMamba. (n) SSLANet.

TABLE II

CLASSIFICATION RESULTS ON THE INDIAN PINES DATASET WITH 10% TRAINING SAMPLES. THE BEST OVERALL RESULTS ARE HIGHLIGHTED IN BOLD

| Class | CNNs | | Transformers | | | | | Mambas | | | | SSLANet |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2D-CNN | HybridSN | DCTN | HiT | SS-TMNet | SSFTTNet | SpectralFormer | MambaHSI | 3DSS-Mamba | $S^2$Mamba | WaveMamba | |
| 1 | 86.10±8.80 | 0.49±1.46 | 53.17±26.58 | 11.71±19.26 | 36.59±23.16 | 70.24±17.31 | 58.78±8.34 | 64.63±12.40 | 18.29±19.94 | 0.98±2.93 | 15.12±21.12 | 78.05±18.09 |
| 2 | 93.15±1.78 | 90.18±3.52 | 82.31±13.57 | 91.53±1.09 | 86.27±2.91 | 86.05±4.19 | 88.00±3.00 | 88.79±2.10 | 85.59±5.64 | 81.04±4.05 | 90.84±2.20 | 93.69±2.11 |
| 3 | 95.03±2.75 | 59.17±21.71 | 90.41±7.47 | 90.51±4.12 | 87.04±5.09 | 91.85±3.27 | 87.39±1.39 | 90.23±4.20 | 61.75±5.96 | 65.01±23.73 | 85.33±7.50 | 95.77±2.09 |
| 4 | 97.84±2.62 | 51.22±37.36 | 94.13±6.37 | 88.87±5.16 | 79.06±11.15 | 86.76±9.38 | 95.59±2.12 | 97.42±1.85 | 78.08±7.91 | 89.15±8.66 | 85.87±10.59 | 96.34±5.14 |
| 5 | 94.53±2.91 | 88.76±8.19 | 88.85±5.16 | 73.31±7.89 | 87.24±11.10 | 89.98±3.83 | 89.08±1.31 | 88.14±5.19 | 65.17±8.22 | 34.30±10.56 | 93.08±2.29 | 97.66±1.44 |
| 6 | 95.22±0.91 | 90.59±22.21 | 93.47±3.83 | 93.88±1.97 | 93.52±1.22 | 93.50±2.51 | 98.48±0.50 | 94.12±2.99 | 94.70±1.36 | 94.09±1.57 | 94.81±2.06 | 96.36±1.45 |
| 7 | 20.80±29.44 | 0.00±0.00 | 8.80±14.18 | 0.00±0.00 | 0.00±0.00 | 35.20±26.16 | 17.60±16.41 | 0.00±0.00 | 0.00±0.00 | 0.00±0.00 | 0.00±0.00 | 18.80±20.00 |
| 8 | 100.00±0.00 | 99.37±1.32 | 98.95±1.62 | 99.98±0.07 | 99.98±0.07 | 100.00±0.00 | 98.95±0.83 | 100.00±0.00 | 98.72±1.33 | 100.00±0.00 | 99.86±0.24 | 99.98±0.07 |
| 9 | 26.67±31.70 | 0.00±0.00 | 51.11±27.98 | 0.00±0.00 | 0.00±0.00 | 7.22±16.86 | 0.00±0.00 | 0.00±0.00 | 10.00±19.05 | 0.00±0.00 | 22.22±20.18 | 76.67±21.05 |
| 10 | 82.09±1.65 | 81.03±6.51 | 73.50±12.40 | 79.13±3.57 | 76.13±3.24 | 77.58±4.89 | 88.56±2.62 | 81.22±1.69 | 71.61±6.04 | 74.72±1.32 | 82.59±2.82 | 84.16±4.87 |
| 11 | 98.49±0.76 | 89.60±6.32 | 90.75±10.56 | 96.09±1.81 | 97.27±1.44 | 96.29±1.92 | 95.89±0.72 | 97.75±0.85 | 90.55±3.04 | 94.79±3.62 | 97.76±0.89 | 98.38±0.83 |
| 12 | 93.84±2.08 | 61.18±30.38 | 85.15±10.71 | 90.88±3.56 | 87.08±8.36 | 88.63±8.21 | 88.03±2.81 | 93.30±4.48 | 76.80±9.26 | 87.08±5.63 | 91.72±5.00 | 87.53±4.79 |
| 13 | 93.08±3.57 | 52.16±43.39 | 83.41±13.26 | 92.70±6.46 | 91.73±5.73 | 86.27±13.65 | 99.19±0.77 | 97.89±1.99 | 85.95±7.76 | 91.46±8.66 | 88.32±12.48 | 93.51±4.07 |
| 14 | 99.63±0.24 | 99.16±1.10 | 97.16±1.75 | 99.50±0.44 | 99.82±0.21 | 97.18±2.77 | 98.89±0.67 | 99.71±0.22 | 98.21±0.93 | 99.40±0.36 | 98.35±1.26 | 98.92±0.81 |
| 15 | 91.67±4.90 | 20.55±21.39 | 83.14±7.07 | 93.83±1.49 | 84.64±7.43 | 87.90±8.19 | 76.57±5.33 | 89.83±3.58 | 62.48±10.56 | 79.65±7.87 | 89.86±4.66 | 93.46±3.03 |
| 16 | 30.60±18.10 | 9.05±27.14 | 14.17±18.11 | 23.57±21.23 | 5.60±16.01 | 20.71±22.44 | 76.19±8.23 | 23.45±21.51 | 17.14±18.31 | 0.00±0.00 | 22.14±24.61 | 35.95±33.41 |
| OA (%) | 94.03±0.46 | 80.64±5.74 | 87.34±4.87 | 90.66±0.68 | 89.45±1.46 | 90.19±0.97 | 91.92±0.47 | 92.12±0.73 | 82.53±1.29 | 83.53±3.35 | 91.51±0.69 | **94.25±0.55** |
| AA (%) | 81.17±3.11 | 55.78±7.96 | 74.28±3.48 | 70.34±1.43 | 69.50±3.11 | 75.96±2.45 | 78.57±1.56 | 75.40±2.04 | 63.44±2.87 | 61.98±3.00 | 72.37±2.03 | **84.08±3.33** |
| $\kappa$ (%) | 93.17±0.53 | 77.75±6.77 | 85.60±5.50 | 89.31±0.78 | 87.89±1.69 | 88.78±1.10 | 90.75±0.55 | 90.99±0.84 | 80.00±1.50 | 81.08±3.87 | 90.29±0.79 | **93.44±0.63** |

*1) Indian Pines Dataset:* On the Indian Pines dataset, 10% of the labeled samples were used for training, with the remaining samples reserved for testing. Table II presents the quantitative comparison results for various methods. Overall, the proposed SSLANet achieved the best performance across multiple evaluation metrics, demonstrating a strong comprehensive performance advantage.

Regarding the different modeling paradigms, CNN-based methods effectively extract local spatial features, but their fixed receptive fields limit the modeling of long-range spectral dependencies. Transformer-based methods extend feature interactions through global attention and provide a complementary modeling paradigm to convolutional architectures. As a representative Transformer comparison, SSLANet improves OA, AA, and Kappa over DCTN by 6.91%, 9.80%, and 7.84%, respectively. Mamba-based methods efficiently model long-range spectral sequences through state-space mechanisms, but more explicit modeling of local spatial structures remains necessary. By combining input-conditioned frequency-domain spectral modulation with multiplicative spatial interaction, SSLANet effectively integrates global spectral responses with local spatial context.

Furthermore, the classification results of each method are visualized in Fig. 2. It can be observed that CNN, Transformer, and Mamba-class methods still exhibit some noise or blurred boundaries in certain regions, whereas SSLANet's classification results align more closely with the true distribution, featuring clear class boundaries and significantly reduced noise, further validating the effectiveness of this method in spectral–spatial joint modeling.

*2) Houston 2013 Dataset:* On the Houston 2013 dataset, experiments were conducted using 1% of the labeled samples for training. As shown in Table III, SSLANet achieves the highest OA, AA, and Kappa among the compared methods under the reported experimental setting. Compared with DCTN, SSLANet obtains improvements of 1.18%, 1.97%, and 1.2% in OA, AA, and Kappa, respectively. In the visualization results shown in Fig. 3, despite the sparse yet locally dense distribution of labeled samples in the scenes, SSLANet still provides the most accurate prediction details. These

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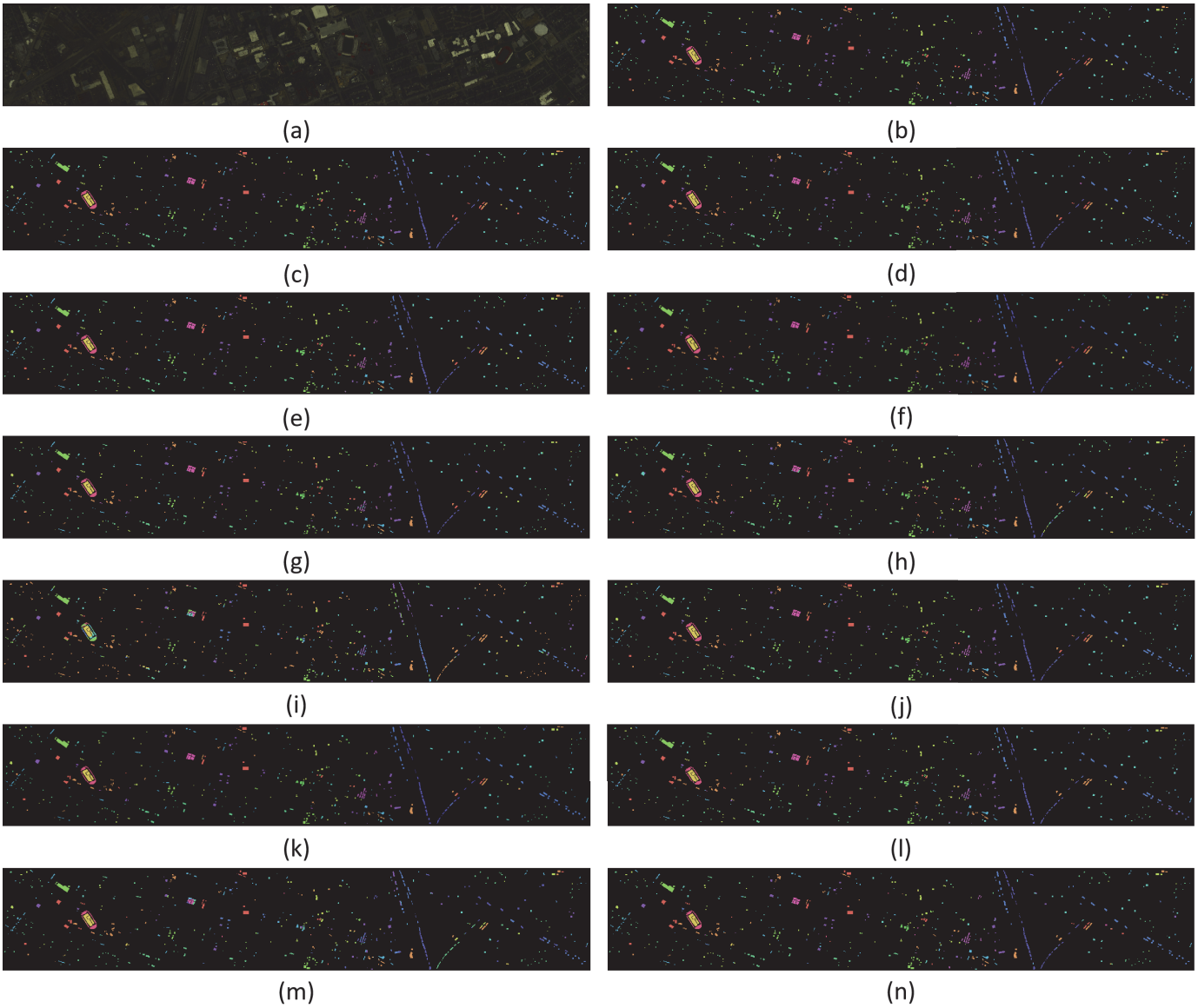



Fig. 3. Visual comparison of the Houston 2013 dataset and classification maps obtained by different methods. (a) RGB composite. (b) Ground truth. (c) 2-D-CNN. (d) HybridSN. (e) DCTN. (f) HiT. (g) SS-TMNet. (h) SSFTTNet. (i) SpectralFormer. (j) MambaHSI. (k) 3DSS-Mamba. (l) $S^2$Mamba. (m) WaveMamba. (n) SSLANet.

outstanding improvements further validate its effectiveness and application potential in urban scene classification under hyperspectral conditions.

*3) WHU-Hi-LongKou Dataset:* On the WHU-Hi-LongKou dataset, the experiments were trained using 0.1% of the labeled samples. The quantitative classification accuracy is shown in Table IV. According to the results, SSLANet achieves the highest OA, AA, and Kappa among the compared methods under the reported experimental setting and also obtains higher classwise accuracies for several challenging categories, including Classes 3, 5, and 9. Compared with 3DSS-Mamba, SSLANet improves OA, AA, and Kappa by 8.42%, 27.94%, and 11.21%, respectively. As shown in Fig. 4, SSLANet produces more spatially continuous classification maps with clearer class regions and fewer visually scattered predictions under the adopted 0.1% training setting. These quantitative and qualitative results support the effectiveness of SSLANet in limited-label hyperspectral classification.

### E. Ablation and Sensitivity Analysis

*1) Patch Size Analysis:* To investigate the impact of spatial receptive-field size, we varied the patch size over {9, 11, 13, 15, 17, 19} on Indian Pines, Houston 2013, and WHU-Hi-LongKou while keeping the remaining settings unchanged. We report the mean and standard deviation of OA, AA, and the Kappa coefficient.

Table V shows that the smaller patches, particularly $9 \times 9$, give the highest numerical accuracy in this sensitivity study. Nevertheless, we retain the prespecified $15 \times 15$ setting for the main benchmark experiments to maintain consistency and avoid posthoc tuning based on test performance. We therefore do not present it as empirically optimal. Instead,

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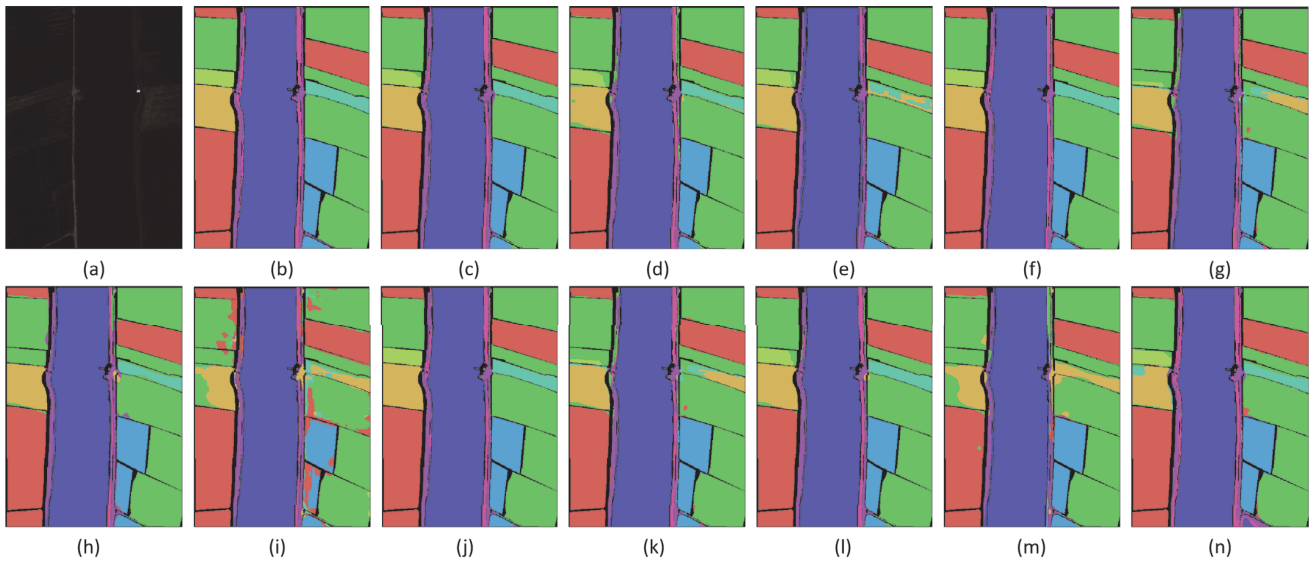



Fig. 4. Visual comparison of the WHU-Hi-LongKou dataset and classification maps obtained by different methods. (a) RGB composite. (b) Ground Truth. (c) 2-D-CNN. (d) HybridSN. (e) DCTN. (f) HiT. (g) SS-TMNet. (h) SSFTTNet. (i) SpectralFormer. (j) MambaHSI. (k) 3DSS-Mamba. (l) S$^2$Mamba. (m) WaveMamba. (n) SSLANet.

TABLE III

CLASSIFICATION RESULTS ON THE HOUSTON 2013 DATASET WITH 1% TRAINING SAMPLES. THE BEST OVERALL RESULTS ARE HIGHLIGHTED IN BOLD

| Class | CNNs | | Transformers | | | | | Mambas | | | | **SSLANet** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2D-CNN | HybridSN | DCTN | HiT | SS-TMNet | SSFTTNet | SpectralFormer | MambaHSI | 3DSS-Mamba | S$^2$Mamba | WaveMamba | |
| 1 | 79.16±12.26 | 77.71±9.70 | 81.39±4.86 | 79.72±4.13 | 82.70±4.32 | 80.86±3.10 | 95.60±0.51 | 84.89±4.59 | 77.00±8.02 | 79.49±1.68 | 85.83±4.05 | 89.10±4.43 |
| 2 | 84.83±7.27 | 88.12±4.13 | 92.34±4.64 | 82.65±9.79 | 81.38±9.56 | 81.35±6.80 | 88.37±3.17 | 82.14±3.89 | 71.51±9.79 | 61.28±18.37 | 75.03±9.84 | 88.89±5.54 |
| 3 | 92.68±7.87 | 95.57±4.36 | 93.20±2.90 | 91.30±3.65 | 94.93±3.75 | 91.84±6.51 | 99.80±0.12 | 92.86±8.34 | 80.09±9.82 | 95.01±4.37 | 96.04±1.44 | 94.87±2.57 |
| 4 | 77.13±9.53 | 72.99±13.02 | 88.77±5.20 | 60.10±7.94 | 84.06±5.31 | 79.80±5.70 | 83.26±2.89 | 85.14±4.77 | 79.02±5.03 | 45.10±25.16 | 82.48±6.04 | 87.09±7.22 |
| 5 | 91.95±6.53 | 98.81±2.01 | 97.54±5.57 | 98.35±2.70 | 97.88±2.12 | 92.58±6.70 | 98.91±0.86 | 98.40±2.10 | 76.08±9.52 | 99.70±0.30 | 92.54±6.81 | 94.98±8.29 |
| 6 | 67.08±13.09 | 65.50±30.82 | 71.68±3.35 | 42.45±14.06 | 60.78±20.66 | 60.93±12.71 | 69.38±3.73 | 60.28±18.15 | 47.42±22.52 | 15.75±15.81 | 67.95±9.60 | 76.24±7.23 |
| 7 | 82.55±7.07 | 33.81±19.26 | 86.69±7.61 | 65.69±10.06 | 90.33±5.84 | 92.09±4.70 | 80.43±2.21 | 85.00±7.77 | 89.73±6.59 | 92.06±4.57 | 86.17±6.53 | 90.99±3.94 |
| 8 | 70.38±7.18 | 50.51±7.84 | 60.92±6.55 | 55.70±8.07 | 65.10±8.76 | 61.10±7.61 | 64.66±3.22 | 76.04±5.68 | 53.30±8.87 | 50.75±12.63 | 50.58±3.59 | 65.39±6.42 |
| 9 | 83.22±8.65 | 42.60±16.04 | 92.96±3.54 | 74.84±7.11 | 82.66±7.98 | 80.41±9.97 | 71.51±4.99 | 80.56±7.14 | 77.59±5.97 | 78.10±5.31 | 74.50±7.58 | 89.82±3.58 |
| 10 | 92.66±5.80 | 75.74±23.66 | 91.53±10.14 | 74.42±15.82 | 76.97±13.93 | 78.81±18.38 | 85.62±2.22 | 80.58±11.07 | 63.65±12.96 | 51.30±19.23 | 75.33±17.57 | 88.02±6.91 |
| 11 | 85.45±7.79 | 52.88±14.14 | 89.88±8.74 | 35.10±17.09 | 79.26±9.20 | 83.11±7.77 | 70.48±5.02 | 87.87±10.77 | 80.38±9.41 | 38.54±20.00 | 85.56±12.84 | 88.00±6.70 |
| 12 | 71.41±11.12 | 81.69±7.49 | 72.92±9.38 | 74.10±12.69 | 79.69±8.61 | 55.53±10.70 | 63.71±2.45 | 74.22±5.04 | 36.05±11.68 | 56.36±9.97 | 51.29±10.29 | 78.11±6.39 |
| 13 | 75.62±11.25 | 2.82±5.82 | 56.40±5.92 | 39.91±25.35 | 66.38±11.18 | 82.91±3.88 | 0.37±1.10 | 73.04±13.40 | 70.56±9.95 | 0.00±0.00 | 78.30±10.61 | 80.69±12.37 |
| 14 | 70.73±17.31 | 67.41±35.18 | 96.84±4.43 | 78.21±12.77 | 82.74±15.36 | 73.51±14.07 | 94.74±1.62 | 95.57±3.61 | 58.80±19.24 | 60.68±20.73 | 80.59±14.12 | 92.10±4.49 |
| 15 | 89.19±11.63 | 98.42±2.79 | 95.22±4.29 | 86.46±10.04 | 95.13±5.26 | 86.17±10.92 | 98.64±1.29 | 92.25±4.08 | 65.64±15.61 | 88.71±15.43 | 68.79±13.18 | 93.45±7.10 |
| OA (%) | 81.86±2.21 | 68.02±3.06 | 85.40±1.48 | 70.48±2.25 | 82.27±3.67 | 79.16±2.56 | 79.67±1.63 | 83.82±2.14 | 69.90±3.03 | 64.52±3.04 | 76.61±1.81 | **86.58±1.51** |
| AA (%) | 80.94±2.62 | 66.97±4.16 | 84.55±1.37 | 69.27±2.34 | 81.33±4.53 | 78.73±2.77 | 77.70±1.65 | 83.25±2.25 | 68.45±3.84 | 60.86±3.68 | 76.73±1.67 | **86.52±1.77** |
| $\kappa$ (%) | 80.38±2.39 | 65.37±3.32 | 84.20±1.60 | 68.03±2.44 | 80.81±3.99 | 77.45±2.77 | 78.01±1.68 | 82.50±2.31 | 67.43±3.30 | 61.51±3.31 | 74.72±1.96 | **85.49±1.64** |

TABLE IV

CLASSIFICATION RESULTS ON THE WHU-HI-LONGKOU DATASET WITH 0.1% TRAINING SAMPLES. THE BEST OVERALL RESULTS ARE HIGHLIGHTED IN BOLD

| Class | CNNs | | Transformers | | | | | Mambas | | | | **SSLANet** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2D-CNN | HybridSN | DCTN | HiT | SS-TMNet | SSFTTNet | SpectralFormer | MambaHSI | 3DSS-Mamba | S$^2$Mamba | WaveMamba | |
| 1 | 97.73±1.23 | 93.42±6.06 | 98.48±0.67 | 99.13±0.39 | 99.20±0.90 | 98.58±1.29 | 99.33±0.27 | 98.84±0.64 | 95.78±2.02 | 97.74±1.58 | 98.40±1.00 | 99.00±0.67 |
| 2 | 84.78±8.77 | 2.47±3.61 | 84.50±4.52 | 81.04±12.64 | 87.76±9.43 | 84.12±9.86 | 77.91±7.39 | 78.59±9.54 | 78.65±7.69 | 51.56±9.47 | 92.62±5.76 | 81.35±11.53 |
| 3 | 31.05±24.67 | 10.97±17.53 | 27.82±32.27 | 2.89±5.69 | 6.28±8.50 | 7.87±20.68 | 0.50±0.77 | 15.49±19.67 | 1.04±2.46 | 0.00±0.00 | 1.32±1.97 | 62.94±14.81 |
| 4 | 98.52±0.64 | 96.51±1.82 | 97.30±1.40 | 98.60±0.73 | 98.90±0.51 | 98.73±0.51 | 87.81±3.41 | 98.57±0.71 | 95.29±2.10 | 99.00±0.54 | 99.25±0.74 | 98.67±0.93 |
| 5 | 34.63±9.40 | 3.61±9.84 | 19.30±12.79 | 18.33±13.98 | 39.49±8.15 | 38.46±28.29 | 14.79±3.01 | 39.69±18.06 | 20.64±22.54 | 0.00±0.00 | 55.34±34.21 | 56.71±18.61 |
| 6 | 87.91±2.49 | 94.06±4.45 | 88.57±5.23 | 95.54±2.36 | 97.42±2.61 | 97.06±3.62 | 74.98±8.72 | 93.71±2.87 | 76.95±8.41 | 81.85±27.25 | 91.31±5.98 | 95.77±3.53 |
| 7 | 99.81±0.17 | 98.40±1.32 | 99.38±0.55 | 99.99±0.01 | 99.99±0.03 | 99.96±0.06 | 99.91±0.12 | 99.96±0.07 | 98.99±1.01 | 100.00±0.01 | 99.90±0.22 | 99.85±0.32 |
| 8 | 68.42±9.91 | 12.48±10.92 | 70.52±10.39 | 76.01±6.15 | 73.98±10.64 | 74.41±15.12 | 60.04±8.56 | 66.58±17.84 | 22.76±16.02 | 52.58±26.33 | 62.52±14.82 | 78.23±8.89 |
| 9 | 68.91±10.64 | 16.96±15.19 | 67.67±17.98 | 43.23±12.71 | 58.99±6.40 | 53.42±10.99 | 31.42±6.75 | 54.78±20.76 | 9.58±7.22 | 0.07±0.20 | 63.79±21.87 | 78.65±11.71 |
| OA (%) | 93.53±0.62 | 84.50±1.23 | 92.85±0.89 | 93.00±0.67 | 94.30±0.79 | 93.84±1.38 | 87.39±0.98 | 93.31±0.83 | 87.21±1.47 | 88.56±1.35 | 94.06±1.46 | **95.63±0.75** |
| AA (%) | 74.64±3.85 | 47.65±2.52 | 72.61±4.83 | 68.31±2.52 | 73.56±2.61 | 72.51±6.25 | 60.74±1.08 | 71.80±4.51 | 55.52±4.03 | 53.64±2.75 | 73.83±6.07 | **83.46±2.83** |
| $\kappa$ (%) | 91.41±0.85 | 78.87±1.74 | 90.54±1.18 | 90.67±0.90 | 92.43±1.07 | 91.80±1.86 | 83.30±1.27 | 91.09±1.13 | 83.02±1.91 | 84.57±1.85 | 92.11±1.95 | **94.23±1.00** |

Table V identifies smaller patches as lighter, higher accuracy alternatives on these data splits.

*2) Depth Sensitivity and Efficiency Analysis:* Table VI reports the classification performance and relative

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TABLE V

PERFORMANCE COMPARISON OF SSLANET UNDER DIFFERENT PATCH SIZES ON THREE HYPERSPECTRAL DATASETS. THE BEST RESULT FOR EACH METRIC IS HIGHLIGHTED IN BOLD

| Patch Size | Indian Pines | | | Houston 2013 | | | WHU-Hi-LongKou | | |
|---|---|---|---|---|---|---|---|---|---|
| | OA (%) | AA (%) | $\kappa$ (%) | OA (%) | AA (%) | $\kappa$ (%) | OA (%) | AA (%) | $\kappa$ (%) |
| 9 | **97.29±0.52** | **92.62±2.31** | **96.90±0.60** | **88.58±1.40** | **88.48±1.49** | **88.46±1.44** | **98.74±0.77** | **88.83±2.37** | **97.52±1.09** |
| 11 | 97.05±0.35 | 89.55±2.25 | 96.64±0.39 | 88.55±1.31 | 88.21±1.32 | 88.44±1.34 | 97.45±0.60 | 87.73±2.30 | 97.13±1.17 |
| 13 | 96.24±0.73 | 88.85±3.58 | 95.71±0.84 | 87.20±1.47 | 87.93±1.57 | 86.06±1.51 | 96.17±0.80 | 86.64±3.59 | 96.76±1.19 |
| 15 | 94.25±0.55 | 84.08±3.33 | 93.44±0.63 | 86.58±1.51 | 86.52±1.77 | 85.49±1.64 | 95.63±0.75 | 83.46±2.83 | 94.23±1.00 |
| 17 | 91.98±1.22 | 81.74±3.14 | 90.86±1.37 | 86.10±1.22 | 86.92±1.23 | 84.87±1.23 | 95.23±0.65 | 82.81±3.63 | 92.53±1.21 |
| 19 | 91.15±1.14 | 77.53±4.41 | 89.89±1.27 | 85.32±1.36 | 86.27±1.44 | 84.02±1.38 | 94.78±0.73 | 81.83±4.29 | 91.94±1.44 |

TABLE VI

DEPTH SENSITIVITY AND COMPUTATIONAL-OVERHEAD ANALYSIS OF SSLANET ON THE INDIAN PINES DATASET. THE COMPUTATIONAL QUANTITIES ARE NORMALIZED WITH RESPECT TO $N = 4$ UNDER THE SAME PROFILING PROTOCOL AND ARE USED ONLY TO CHARACTERIZE THE COMPUTATIONAL-OVERHEAD TREND WITH INCREASING DEPTH. THE BEST OA, AA, AND KAPPA VALUES ARE HIGHLIGHTED IN BOLD

| $N$ | OA (%) | AA (%) | Kappa (%) | Relative Params | Relative FLOPs | Relative Latency |
|---|---|---|---|---|---|---|
| 1 | 94.20±0.28 | 84.29±1.94 | 93.39±0.32 | 0.314 | 0.292 | 0.332 |
| 2 | 94.14±0.37 | 84.60±2.64 | 93.31±0.42 | 0.543 | 0.528 | 0.551 |
| 3 | 94.11±0.27 | 83.53±2.84 | 93.28±0.31 | 0.771 | 0.764 | 0.770 |
| 4 | 94.25±0.55 | 84.08±3.33 | 93.44±0.63 | 1.000 | 1.000 | 1.000 |
| 5 | **94.28±0.32** | **85.85±1.52** | **93.47±0.36** | 1.229 | 1.236 | 1.208 |
| 6 | 94.01±0.54 | 81.97±2.60 | 93.17±0.61 | 1.457 | 1.472 | 1.430 |

computational overhead for $N = 1$ to $N = 6$. The OA remains stable at 94.20%, 94.14%, and 94.11% for $N = 1$ to $N = 3$, respectively. Increasing $N$ from 4 to 5 improves OA only from 94.25% to 94.28%, a gain of 0.03%, indicating performance saturation. At $N = 6$, OA, AA, and Kappa decrease to 94.01%, 81.97%, and 93.17%, respectively, suggesting that further stacking within the fixed $15 \times 15$ neighborhood may introduce redundant or mixed contextual information.

The computational overhead exhibits a clearer increasing trend. Relative to $N = 4$, the parameters, FLOPs, and latency at $N = 5$ increase to 1.229, 1.236, and 1.208, respectively, for only a 0.03% OA gain. At $N = 6$, these quantities further increase to 1.457, 1.472, and 1.430, while the classification performance declines. Thus, computational cost continues to grow with depth, whereas accuracy improvement does not scale accordingly.

Across $N = 1$ to $N = 5$, the mean OA values lie within a narrow 0.17% range, so this experiment does not establish $N = 4$ as uniquely more accurate than the shallower alternatives. In particular, $N = 1$ reaches 94.20% OA with only 0.314× the parameters, 0.292× the FLOPs, and 0.332× the latency of $N = 4$, making it a viable lightweight operating point. We retain $N = 4$ because it is the prespecified common architecture used for the main benchmark results, not because it maximizes test accuracy. The depth study is therefore presented as a sensitivity and deployment tradeoff analysis rather than a posthoc model-selection procedure.

*3) Temperature Sensitivity Analysis:* To evaluate sensitivity to the temperature initialization, we set $T_0$ to $\{0.05, 0.1, 0.5, 1.0, 2.0, 5.0\}$ on the Indian Pines dataset while fixing the patch size at $15 \times 15$, the training ratio at 10%, and all other experimental settings. For each setting, all ASB layers are initialized with the same $T_0$, after which each layer independently learns its own temperature according to (19).

As shown in Table VII, OA ranges from 94.25% to 94.69% as $T_0$ varies from 0.05 to 5.0. The highest OA and Kappa are obtained at $T_0 = 0.5$, while the highest AA is obtained at $T_0 = 5.0$. The modest differences among these settings indicate that the classification performance is relatively robust to temperature initialization. We therefore retain $T_0 = 1.0$ as a simple and stable default rather than because it yields the highest numerical result.

The learned temperatures generally remain close to their respective initializations. For example, with $T_0 = 1.0$, the learned temperature is $0.9847 \pm 0.0065$, with an overall range of 0.9648–1.0102. Similar trends are observed for the other initializations. These results show that the temperature remains numerically stable during training while preserving independent layerwise adjustment of the soft-mask sharpness.

*4) Module Importance Analysis:* Table VIII reports the ablation results on the Indian Pines dataset. The full model achieves 94.25% OA, 84.08% AA, and 93.44% Kappa. We therefore interpret these mean differences as descriptive incremental contributions and do not use them alone to claim statistically significant standalone improvements. ASB primarily provides lightweight frequency-domain modulation, whereas ICB accounts for a larger share of the model capacity. The complete model, which achieves 94.25% OA with 3.97M parameters and 1.00G FLOPs, should therefore be viewed in terms of its OA–complexity tradeoff.

Removing ASB and ICB reduces the mean OA to 94.08% and 94.02%, respectively. Removing the spectral modulation or energy-guided filtering branch within ASB, as well as replacing FFT with DCT or a wavelet transform, produces the same downward trend. Similarly, removing the multiplicative

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TABLE VII

TEMPERATURE SENSITIVITY ANALYSIS OF SSLANET ON THE INDIAN PINES DATASET. LEARNED $T$ REPORTS THE MEAN AND RUNWISE STANDARD DEVIATION OF THE LAYER-AVERAGED LEARNED TEMPERATURES OVER TEN INDEPENDENT RUNS, WHILE LEARNED $T$ RANGE GIVES THE MINIMUM AND MAXIMUM OVER ALL LEARNED TEMPERATURES ACROSS THE FOUR ASB LAYERS AND TEN RUNS. THE BEST OA, AA, AND KAPPA VALUES ARE HIGHLIGHTED IN BOLD

| $T_0$ | OA (%) | AA (%) | Kappa (%) | Learned $T$ | Learned $T$ Range |
|---|---|---|---|---|---|
| 0.05 | 94.61±0.41 | 84.53±4.18 | 93.86±0.55 | 0.0501±0.0004 | 0.0484–0.0521 |
| 0.1 | 94.47±0.58 | 83.60±3.24 | 93.68±0.74 | 0.1001±0.0013 | 0.0959–0.1050 |
| 0.5 | **94.69±0.34** | 84.66±2.94 | **93.97±0.47** | 0.4924±0.0026 | 0.4757–0.5033 |
| 1.0 | 94.25±0.55 | 84.08±3.33 | 93.44±0.63 | 0.9847±0.0065 | 0.9648–1.0102 |
| 2.0 | 94.39±0.63 | 84.35±3.59 | 93.60±0.80 | 1.9817±0.0068 | 1.9345–2.0091 |
| 5.0 | 94.50±0.31 | **84.97±3.59** | 93.72±0.43 | 4.9807±0.0092 | 4.9440–5.0199 |

TABLE VIII

ABLATION STUDY OF SSLANET ON THE INDIAN PINES DATASET. OA, AA, AND KAPPA ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER 10 INDEPENDENT RUNS, AND ΔOA IS MEASURED RELATIVE TO THE FULL SSLANET

| Component | Variant | OA (%) | AA (%) | $\kappa$ (%) | ΔOA |
|---|---|---|---|---|---|
| Full model | **Full SSLANet** | **94.25±0.55** | 84.08±3.33 | **93.44±0.63** | – |
| ASB | w/o ASB | 94.08±0.65 | 83.35±2.74 | 93.15±0.74 | −0.17 |
| | w/o spectral modulation branch | 94.13±0.63 | 82.74±1.56 | 93.28±0.72 | −0.12 |
| | w/o energy-guided filtering branch | 94.12±0.55 | 83.13±3.84 | 93.27±0.62 | −0.13 |
| | hard energy mask instead of soft mask | 94.17±0.55 | 83.57±1.92 | 93.13±0.63 | −0.08 |
| | FFT → DCT | 94.10±0.74 | 83.08±4.07 | 93.14±0.85 | −0.15 |
| | FFT → wavelet transform | 94.01±0.72 | 83.44±2.23 | 93.34±0.82 | −0.24 |
| ICB | w/o ICB | 94.02±0.42 | 83.86±2.10 | 93.25±0.48 | −0.23 |
| | ICB → standard 3×3 convolution | 94.10±0.54 | **84.11±3.06** | 93.03±0.62 | −0.15 |
| | w/o multiplicative interaction | 94.17±0.56 | 83.66±2.93 | 93.36±0.64 | −0.08 |
| Feature aggregation | w/o dual-branch feature aggregation | 93.08±0.83 | 82.25±3.01 | 92.25±0.67 | −1.17 |

TABLE IX

PERFORMANCE ANALYSIS OF SSLANET UNDER DIFFERENT TRAINING RATIOS ON THREE HYPERSPECTRAL DATASETS. THE BEST RESULT FOR EACH METRIC IS HIGHLIGHTED IN BOLD

| Training Ratio | Indian Pines | | | Houston 2013 | | | WHU-Hi-LongKou | | |
|---|---|---|---|---|---|---|---|---|---|
| | OA (%) | AA (%) | $\kappa$ (%) | OA (%) | AA (%) | $\kappa$ (%) | OA (%) | AA (%) | $\kappa$ (%) |
| 1% | 78.54±0.96 | 61.92±2.42 | 75.23±1.13 | 86.58±1.51 | 86.52±1.77 | 85.49±1.64 | **98.60±0.22** | **95.82±0.73** | **98.16±0.29** |
| 5% | 92.68±0.41 | 79.87±3.31 | 91.63±0.46 | 96.61±1.31 | 95.76±1.39 | 96.34±1.33 | 98.48±0.12 | 94.77±0.32 | 98.01±0.16 |
| 10% | 94.25±0.55 | 84.08±3.33 | 93.44±0.63 | 97.81±1.16 | 97.15±1.12 | 97.63±1.18 | 98.54±0.14 | 95.09±0.49 | 98.08±0.18 |
| 20% | **95.78±0.26** | **87.34±2.58** | **95.18±0.29** | 98.57±1.10 | 98.26±1.14 | 98.46±1.11 | 98.51±0.14 | 94.93±0.48 | 98.04±0.18 |
| 30% | 95.45±0.32 | 85.90±3.92 | 94.80±0.37 | **98.65±1.18** | **98.30±1.27** | **98.54±1.19** | 98.59±0.15 | 95.65±0.61 | 98.15±0.20 |

interaction within ICB also reduces OA. Although replacing ICB with a standard $3 \times 3$ convolution yields a marginally higher AA, its OA remains below that of the full model. Overall, these results support the complementary roles of ASB and ICB in frequency-domain modeling and spatial–channel interaction.

The most pronounced degradation occurs when the dual-branch feature aggregation is removed. Compared with the full model, OA decreases from 94.25% to 93.08%, AA from 84.08% to 82.25%, and Kappa from 93.44% to 92.25%, corresponding to reductions of 1.17, 1.83, and 1.19 percentage points, respectively. By combining mean contextual statistics with salient discriminative responses, this aggregation provides a more complete final representation. Therefore, the observed degradation reflects an information bottleneck at the final readout stage rather than indicating that feature aggregation is more important than ASB or ICB in the overall architecture.

*5) Classwise Frequency-Energy Analysis:* To examine whether the learned latent frequency basis exhibits class-dependent differences, Fig. 5 visualizes the classwise mean normalized frequency-energy profiles extracted from the first ASB, covering 8427 valid-center test samples from the fixed Indian Pines test split and 129 nonredundant latent frequency indices. The profiles exhibit nonuniform class-dependent differences: the mean interclass variances over indices 0–42, 43–85, and 86–128 are 0.0676, 0.0792, and 0.0926, respectively, with a maximum of 0.6807 at index 102. These results indicate class-dependent energy distributions in the learned latent frequency representation, providing empirical support for input-conditioned frequency modulation. However, the latent indices do not correspond to the physical wavelengths of the original hyperspectral bands, and the observations do not imply that the frequency domain is intrinsically superior to the spatial–spectral representation or that any class is distinguishable only in the frequency domain.

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TABLE X

COMPUTATIONAL COMPLEXITY AND EFFICIENCY COMPARISON ON INDIAN PINES. FPS IS MEASURED ON AN NVIDIA GEFORCE RTX 4090 (24 GB) USING 15 × 15 PATCHES, BATCH SIZE 100, DEFAULT PYTORCH INFERENCE PRECISION, SIX WARM-UP MINIBATCHES, CUDA SYNCHRONIZATION, AND ONE COMPLETE TEST-LOADER PASS. TRAINING TIME IS THE FIXED 30-EPOCH BENCHMARK TRAIN LOOP AND IS SEPARATE FROM THE 200-EPOCH CLASSIFICATION SCHEDULE. THE BEST RESULT IN EACH COLUMN IS HIGHLIGHTED IN BOLD

| Method | Params (M) | FLOPs (G) | FPS ↑ | Training Time (s) ↓ | OA (%) ↑ |
|---|---|---|---|---|---|
| 2D-CNN | 1.98 | 0.16 | 2576.80 | **13.74** | 94.03±0.46 |
| HybridSN | 4.32 | 1.61 | **3892.38** | 32.11 | 80.64±5.74 |
| DCTN | 44.17 | 4.72 | 1190.30 | 126.04 | 87.34±4.87 |
| HiT | 27.58 | 1.42 | 1554.76 | 55.95 | 90.66±0.68 |
| SSFTTNet | **0.02** | 0.12 | 2220.17 | 17.29 | 90.19±0.97 |
| MambaHSI | 0.39 | 0.11 | 1265.18 | 54.10 | 92.12±0.73 |
| 3DSS-Mamba | 2.51 | 0.23 | 1371.15 | 51.70 | 82.53±1.29 |
| $S^2$Mamba | 0.11 | **0.04** | 2290.50 | 19.70 | 83.53±3.35 |
| WaveMamba | 1.19 | 0.28 | 1366.80 | 49.13 | 91.51±0.69 |
| **SSLANet** | 3.97 | 1.00 | 2283.28 | 26.24 | **94.25±0.55** |

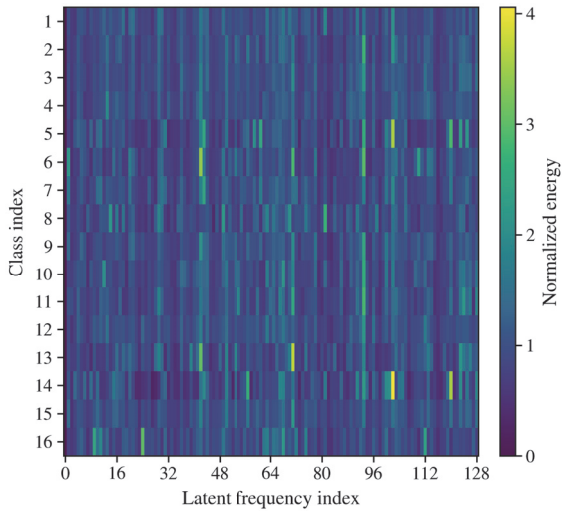



Fig. 5. Classwise mean normalized latent frequency-energy distributions extracted from the first ASB block on the fixed Indian Pines test split. Colors indicate the spatially averaged normalized energy at each latent-frequency index.

*6) Training Ratio Analysis:* To evaluate the robustness of SSLANet to label availability, we vary the training sample ratio from 1% to 30% on the Indian Pines, Houston 2013, and WHU-Hi-LongKou datasets while keeping all other settings fixed, and report OA, AA, and Kappa in Table IX. Overall, the classification performance improves as more labeled samples become available. SSLANet nevertheless maintains favorable performance under low training ratios on Indian Pines and Houston 2013 and remains relatively stable across different ratios on WHU-Hi-LongKou, demonstrating its adaptability to varying labeled-sample conditions.

### *F. Complexity and System Efficiency Analysis*

Table X evaluates computational efficiency in terms of parameters, FLOPs, FPS, and training time under the Indian Pines setting. All measurements are conducted on the same hardware platform under the Indian Pines benchmark setting, with patch size 15 and batch size 100. FPS uses default PyTorch inference precision without mixed-precision acceleration. Six initial minibatches are used for GPU warm-up, CUDA is synchronized immediately before and after timing, and throughput is computed over one complete pass through the test loader, including DataLoader iteration and host-to-device transfer. For training time, we report the fixed 30-epoch benchmark training time on the Indian Pines training split, covering the train loop including batch iteration, forward propagation, backward propagation, and optimizer updates, while excluding validation and DataLoader construction.

Architecturally, ASB models global latent-frequency responses through a 1-D FFT along the channel dimension, while ICB realizes factorized multiplicative spatial–channel interactions using $1 \times 1$ and $3 \times 3$ convolutions without explicitly constructing a dense bilinear interaction tensor. As shown in Table X, SSLANet uses 3.97M parameters and 1.00G FLOPs, compared with 44.17M and 4.72G for DCTN, while achieving 94.25% OA. Its throughput of 2283.28 frames/s is comparable to the 2220.17 frames/s of SSFTTNet, with a 4.06-percentage-point OA improvement, although SSLANet requires greater model capacity and computation. These results therefore characterize an accuracy–efficiency tradeoff rather than a universal efficiency ranking.

## V. CONCLUSION AND FUTURE WORK

This article proposes a SSLANet, which effectively captures the spectral–spatial coupling characteristics of hyperspectral data through the joint design of frequency-domain spectral modeling and spatial interactive learning. Specifically, the ASB enhances discriminative spectral information from a frequency-domain perspective, while the ICB strengthens the representation of spatial structures through factorized multiplicative spatial–channel interaction. These two components are jointly optimized via a residual mechanism and combined with dual-branch feature aggregation to summarize complementary mean and peak responses before classification. Experimental results on multiple public datasets demonstrate that SSLANet achieves competitive classification performance with consistent results across repeated evaluations, supporting

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the effectiveness and generalization capability of the proposed frequency–spatial learning framework.

Future work will focus on model lightweight design, refined frequency-domain modeling, and multimodal remote sensing data fusion to further improve the model's computational efficiency, expressive power, and practical application value.

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